

7	(a)	either	the value of steady / constant voltage	M1	
			that produces same power (in a resistor) as the alternating voltage	A1	[2]
		or	if alternating voltage is squared and averaged	(M1)	
			the r.m.s. value is the square root of this averaged value	(A1)	

(b)	(i)	220 V	A1	[1]
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(ii)	156 V	A1	[1]
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(iii)	60 Hz	A1	[1]
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(c)	power = V_{rms}^2 / R	C1	
	$R = 156^2 / 1500$		
	= 16 Ω	A1	[2]

8	(a)	(i)	number = $(5.1 \times 10^{-6} \times 6.02 \times 10^{23}) / 244$	C1	
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